

THE PRODUCTION GAME

MET IIA – [Friday 13 March 2026](#)

Venue: The Design Studio, IfM

TIMETABLE

Collection Deadline: You **MUST** collect your starter pack of game materials from The Teaching Office following Lecture 6 on **Wednesday 4 March 2026**.

Submission Deadline 1: The pre-game team report **MUST** be submitted **via Moodle** before **1300hrs on Friday 13 March 2026**. See appendix for further details.

Schedule for the Production Game on Friday 13 March 2026

12:00 Design studio closed for game set up

13:00 Design studio opens (you are not allowed to enter the room before this)

13:05 Briefing

13:30 Trading commences

15:30 Trading ceases
All materials to controllers.
Companies reflect on the game and prepare short report-backs.
Controllers calculate performance.

16:30 (approx.) Discussion
Each company presents on its strategy and performance (10 mins each)

17:15 (approx.) Close

Submission Deadline 2: The final individual coursework **MUST** be submitted via Moodle before **1200hrs on Friday 20 March 2026**. See appendix for further details.

TEAM ALLOCATIONS

For the Production Game, the MET IIA class is divided into **FOUR** teams, each of which constitutes a company. Before the game, it is very important that you meet in your teams (listed below) and plan how you will tackle the Game. Once you get into the game itself you will be very busy, and planning will be extremely difficult. Moreover, each team is expected to submit a pre-game report that outlines your team's strategy for the game.

Company A	Company B
Daniel Bohm	Bryan Bukoh
Ossian Finch	Reiko Fujiyoshi
James Goh	Annabel Gray
Amy Jackson	Keshav Kedia
Judy Tang	Praewa Taylor
Jay Yu	

Company C	Company D
Isobel Carson	Oliver Muthu
Amelie Gadsby	Owen Dyson
Leo Hammett	Lithikga Gobi
Gauthami Vireswer	Caragh Haspel
Rick Suzuki	Ran Xin

THE PRODUCTION GAME

Introduction

The Production Game is a simulation of a manufacturing operation. Small companies (teams) are required to manufacture simple paper-based products based on orders from a market place. Products which meet the required standards of quality and delivery are purchased by the market. Unacceptable products are rejected.

At the start of the game, each team is provided with a starter pack of equipment and materials. You may purchase more capital equipment and raw materials as the game goes on. The game is cashless; the controllers maintain a record of debits and credits to your company's account. However, each company is strongly advised to keep its own records of payments and receipts in order to monitor its performance during the game and to resolve any potential differences between the controllers and the team. In the case of discrepancies, the controllers' judgement is final. Equipment and materials available for purchase include:

1.	12" rulers	£300
2.	5mm stencils	£400
3.	10mm stencils	£500
4.	Pens	£300
5.	Pencils	£100
6.	Rubbers	£300
7.	Paper - white (per sheet)	£10
8.	Paper - coloured (per sheet)	£20
9.	Envelopes (each)	£10

Extra materials or equipment must be ordered using the special procurement forms provided (a specimen form is included in this document). There is a 10-minute lead-time on all purchases. The maximum value of goods ordered in a single order must not exceed £1000, and you must leave a minimum of 5 mins between placing consecutive orders. At the end of the game capital equipment and unused consumables will be valued at 30% of their original purchase price.

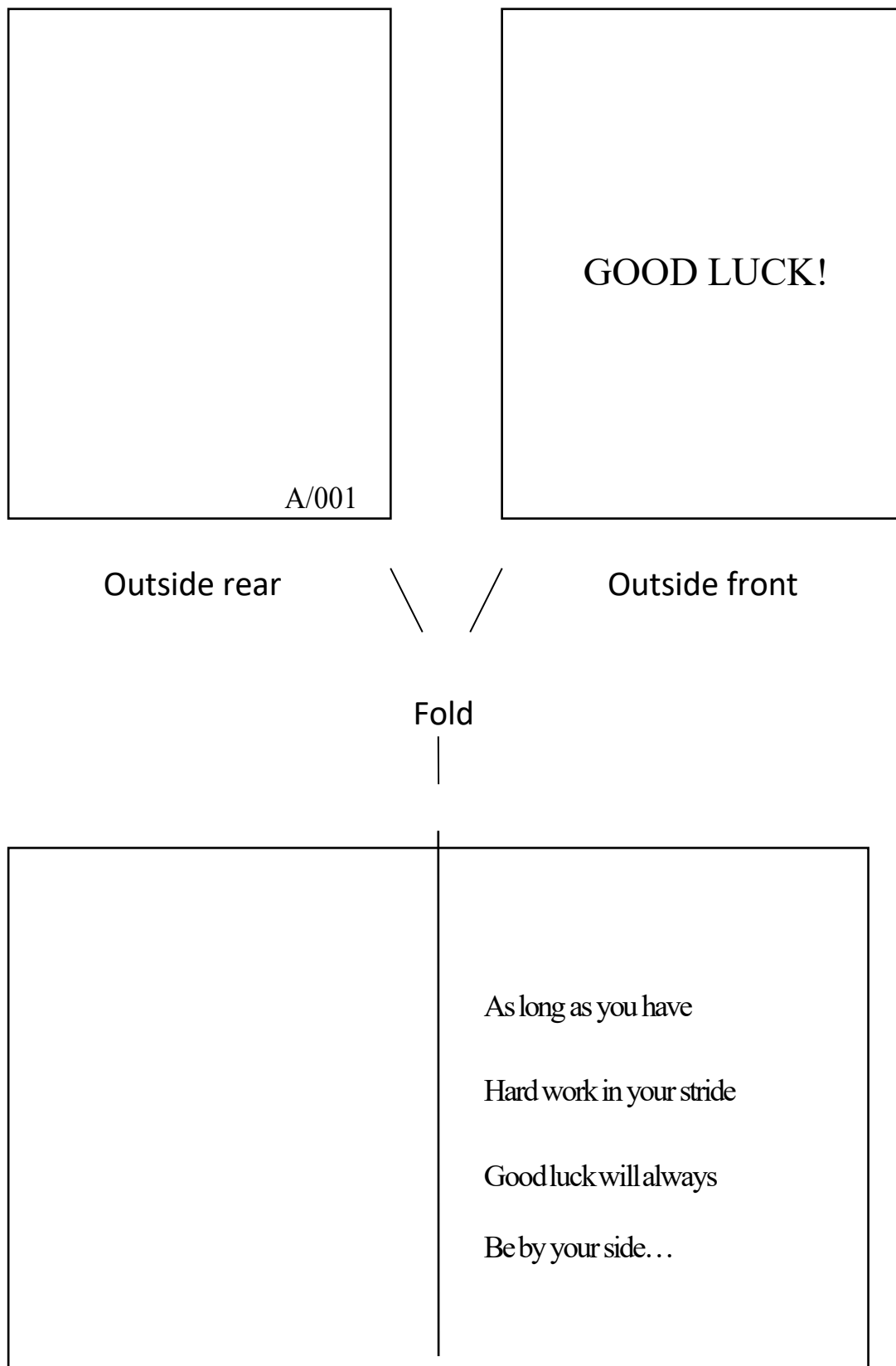
Products and Company Operations

Each team produces greeting cards in a range of colours and sizes (A5, A6 and A7), with the occasion stencilled on the front of the card and an appropriate verse written on the inside. The market will buy at a declared price any order that is complete, manufactured to specification and delivered at or before the required time. Some cards (for example Christmas) may be ordered to a delivery date, that is a time in the game when they must be ready. Other cards, such as birthdays, may be open delivery and can be delivered at any time up to the end of play. There will be special orders, which embody some unusual requirement of size, shape or occasion and will carry a premium price.

Paper is supplied as standard commercial A4 sheets. To make an A5 card fold an A4 sheet in half along the short dimension. The short dimension of the A4 is now the long dimension of the A5 - an A5 is hence half an A4. An A6 is made by folding an A5 sheet in half and an A7 by folding an A6 in half. The cards are formed by folding only. No cutting is required.

The format of a finished card is shown in Figure 1. Seen from the front the fold runs down the long left-hand side, with the occasion stencilled across the middle. The specification states that this should be centred in both the top-down and left-right dimensions. The manufacturing team's identity and the sales order number is to be written on the rear bottom right-hand corner next to the fold. Inside the card the verse must rhyme, be neatly set out, legible and not enter the margins given in the specifications. Verse must be written on the right-hand inside page.

Figure 1: Format of an A7 Good Luck card



Pencil sharpeners will be available with the procurement controllers, but they cannot be borrowed. Scheduling of maintenance might turn out to be critical.

Inter-company sales are allowed (raw materials, finished goods, outsourcing). Fill out the INTER-COMPANY TRANSACTION FORM and submit to the procurement controller – these can be obtained free of cost. Exchanges (bartering) need not be recorded.

The Market

The marketplace is represented by an ‘order board’. Orders are placed on the board by the controllers. Each order specifies the number and type of products required, the time by which they are required and the price that will be paid. As described above, some orders will specify ‘open’ delivery and will be accepted at any point of the game’s duration. A team accepting a particular order undertakes to meet particular delivery and quality requirements with the game’s controllers. For an order to be valid it must have a delivery deadline specified, *and be signed by one of the controllers*, to whom the finished product must be delivered. Orders that are taken on but not completed by the deadline are rejected, and due to a penalty clause render the company liable for 20% of the value of the order. Orders vary in terms of their size, complexity, and delivery deadlines and of course price. Generally, the more demanding the order is, the more money it earns for the company completing it successfully. Sample orders from previous games are shown in Table 2. These accurately represent the nature of the market, and may be used to assess the relative attractiveness of orders.

Table 2: Typical Market Conditions

The data given below are drawn from a recent market research report on the market in which you will be operating. A reputable company produced the report, and it is safe to assume that it fairly represents the conditions under which you will be operating.

QUANTITY	COLOUR	SIZE	LEAD TIME	LINES OF VERSE	OCCASION	PRICE
4	White	A5	15mins	6	Christmas	£106
12	Green	A6	25mins	4	New Year	£504
8	Pink	A5	Open	6	Wife’s birthday	£232
4	White	A6	25mins	4	Father’s birthday	£84
12	Salmon	A5	20mins	6	Get Well Soon	£768
8	Yellow	A6	30mins	4	Baby Girl	£280
6	Blue	A5	15mins	6	Baby Triplets	£212
12	White	A6	15mins	4	Good Luck	£516
8	Green	A5	25mins	6	St David’s Day	£320
4	Pink	A6	Open	4	Mother’s Day	£124
12	White	A5	25mins	6	Examination Pass	£252
8	Salmon	A6	20mins	4	Marriage	£440
12	Green	A7	25mins	2	Pregnancy	£504
4	Yellow	A5	30mins	6	New Job	£144
12	Blue	A6	15mins	4	18 Birthday	£1032
8	White	A5	15mins	6	Driving Test Pass	£280
4	Green	A6	25mins	4	New Home	£168
8	Yellow	A7	30mins	2	Passover	£280
12	Pink	A5	Open	6	Easter	£324
8	White	A6	25mins	4	Silver Wedding	£160

Orders are placed using a standard 'SALES ORDER AND DELIVERY NOTICE (SODN)', a specimen of which is appended.

Delivery

All the cards in a given order must be delivered together packed in an envelope with the original SODN. Each order must be delivered in a separate envelope, marked on the outside with the identity of the manufacturer, which must also be written on the SODN. Neither part deliveries nor envelopes containing more than one sales order will be accepted, and no company will be paid for unidentified consignments.

If anything in the order is not up to specification, the whole order will be returned to its makers together with an indication of why the order has been rejected. That order is then lost, although salvageable materials may be re-used in later orders as long as they meet the specification.

Organization and Strategy

The teams may choose any form of organization they consider appropriate. Good organization is crucial to commercial success in the game, and careful thought on this will pay dividends. The process consists of 10 identifiable activities:

- selecting orders
- reducing sheets to size/folding sheets
- verse composition
- copying verses
- copying occasion (front cover)
- copying manufacturer's identity and order number (back cover)
- packing
- completing SODN
- labelling envelope
- delivery

There are five administrative functions:

- identifying and selecting orders (sales/marketing)
- estimating and ordering raw materials (stock control)
- processing and organization of manufacturing (production control)
- cost and revenue calculations (accounting)
- record keeping as required

You need to think hard about an appropriate division of labour. Are you going to go for a functional or a product-based layout? What coordinating (integrating) mechanisms are you going to use? What strategy are you pursuing? What does this imply for the structure you adopt? How are decisions going to be made? And so on.

Trading runs for two hours. Up to two short breaks (of 5-10 mins each) may be given at any time during the game, either with or without prior notice. During such a break, all production will cease on as-it-is basis and will resume at the end of the break. This may cause disruption to your operations but such breaks, if given, may be utilised for reviewing team performance and reconfiguring strategies to align with the objectives of the game. At the end of this time the performance of each company (in terms of turnover, profit and loss, delivery and quality) will be calculated and compared. Each company will be required to make a short presentation on their performance during the game.

APPENDIX

Contains following:

- Sales Order and Delivery Notice (SODN) (specimen);
- Specifications (specimen);
- The Starter Pack contents (specimen);
- Procurement Form (specimen);
- Final Accounts Form (specimen);
- Inter-company Transaction Form (specimen); and
- 3P4/3P5 Coursework Assignment.

SALES ORDER AND DELIVERY NOTICE (SODN)
(to be retained by controllers after delivery)
(v2.0, Feb 2015)

Sales order number:

Company

Occasion	Quantity	Colour	Size	Number of lines of verse	Lead time	Price

Time order granted

Accept

Comments

Time due

Reject

Controller signature

..... ..

SPECIFICATIONS

(v2.0, Feb 2015)

1. Text inside and on the rear of the card to be neatly written.
2. All lines of text to be perfectly horizontal.
3. Title to be centred, both horizontally and vertically and stencilled on front cover. A5 cards must use 10mm stencils; A6 can use 5mm or 10 mm; A7 must use 5 mm stencils.
4. Verse text must not exceed the following margins:

Card	Left	Right	Top	Bottom
A5	30mm	30mm	40mm	40mm
A6	20mm	20mm	30mm	30mm
A7	10mm	10mm	20mm	20mm

5. Verses must rhyme.
6. Text to be written using pens provided with no smudges.
7. All cards in a batch to have the same verse.
8. No pencil marks to be visible.
9. Closed folds: - to be at the top of A6 and A7 cards
- to be on the frontispiece of A7 cards.
10. Manufacturers identity and sales order number to be written on the back of every card in the bottom right-hand corner next to the fold.
11. Products to be delivered in an envelope, marked with the company's ID and the sales order number.
12. Neither part-deliveries nor envelopes containing more than one sales order will be accepted.
13. Missed deadlines or non-delivery mean the loss of that order plus a penalty charge of 20% of the value of the order.
14. No equipment and consumables to be used other than those bought from the game's controllers. However, teams are allowed to bring pre-prepared templates.

IF IN DOUBT – CONSULT THE CONTROLLERS

THE STARTER PACK CONTENTS

(v2.0, Feb 2015)

All companies are issued with a starter pack containing following items (see Figure 2):

1 12" Ruler		£ 300
1 5-mm stencil		£ 400
1 10-mm stencil		£ 500
1 Pen		£ 300
1 Pencil		£ 100
1 Rubber		£ 300
10 Envelopes	@ £10	£ 100
10 sheets of white	@ £10	£ 100
10 sheets of salmon	@ £20	£ 200
10 sheets of pink	@ £20	£ 200
10 sheets of blue	@ £20	£ 200
10 sheets of green	@ £20	£ 200
10 sheets of yellow	@ £20	£ 200
TOTAL VALUE:		£3100

Figure 2: A typical starter pack



Extras of the above items are available at cost during the game.

Each pack also contains the following material which is offered free of charge:

- 3 SPECIFICATIONS lists
- 20 PROCUREMENT forms
- A FINAL ACCOUNTS form
- 5 INTER-COMPANY TRANSACTION forms

Extras of these are available free-of-charge during the game.

PROCUREMENT FORM

(v2.0; Feb 2015)

To be used for ordering equipment and consumables: To be retained by controllers on final delivery of the order.

ALL ORDERS HAVE A LEAD TIME OF 10 MINUTES AND A MAXIMUM VALUE OF £1000. YOU MUST LEAVE A MINIMUM OF 5 MINUTES BETWEEN PLACING CONSECUTIVE ORDERS.

Company:

Capital Equipment

	<u>Quantity</u>	Price	Total
12" rulers		£300	
5 mm stencils		£400	
10 mm stencils		£500	
Pens		£300	
Pencils		£100	
Rubbers		£300	

Consumables

Paper - white (per sheet)		£10	
Paper - coloured (p/sheet)			
Sheets of salmon		£20	
Sheets of pink		£20	
Sheets of blue		£20	
Sheets of green		£20	
Sheets of yellow		£20	

Envelopes (each)		£10	
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TOTAL PAYABLE:

Time order placed:

Earliest collection time:

FINAL ACCOUNTS FORM

(V2.0; Feb 2015)

Company Name

INCOMINGS/ASSETS

Total value of sales £.....

Assets at end of game

	Quantity		Total
Ruler		@£300	£.....
5 mm stencil		@£400	£.....
10 mm stencil		@£500	£.....
Pen		@£300	£.....
Pencil		@£100	£.....
Rubber		@£300	£.....
Envelopes		@£ 10	£.....
Sheets of white		@£ 10	£.....
Sheets of salmon		@£ 20	£.....
Sheets of pink		@£ 20	£.....
Sheets of blue		@£ 20	£.....
Sheets of green		@£ 20	£.....
Sheets of yellow		@£ 20	£.....

Total Assets £.....

Assets at end of game (30% of total) £.....

TOTAL INCOMINGS/ASSETS £.....

OUTGOINGS/LIABILITIES

Starter pack £ 3100

Materials purchased £.....

Fines and penalties £.....

TOTAL OUTGOINGS/LIABILITIES £.....

PROFIT/LOSS £.....

Number of rejected orders

INTER-COMPANY TRANSACTION FORM

(v2.0; Feb 2015)

REF NO (TO BE ASSIGNED BY CONTROLLER):

SOLD BY (COMPANY NAME):

BOUGHT BY (COMPANY NAME):

VALUE OF TRANSACTION (£):

NAME OF REPRESENTATIVE (SELLER):

NAME OF REPRESENTATIVE (BUYER):

3P4/3P5 Coursework Assignment 2026

*Pre-game (team) report to be submitted via Moodle before 1300hrs on Friday 13 March 2026.
Final (individual) report to be submitted via Moodle before 1200hrs on Friday 20 March 2026.*

The Production Game typically provides a rich array of first-hand experience with which to think about the issues involved in the organization and control of manufacturing systems. The purpose of the assignment is to allow you to draw out the lessons that emerge from the Game, and to think about how principles covered in 3P4 and 3P5 apply to the Game.

You are required to submit two reports:

1. Pre-game report (team): Each team is required to submit a report of a maximum 2,000 words in length, outlining the following:
 - What is your strategy?
E.g. What products/orders will you target? What is your expected overall capacity utilisation? How flexible will you be? What is your budgeted profit?
 - Production layout, resource allocation and operations:
E.g. Are you going to go for a functional or a product-based layout? How are your resources going to be allocated? Where are the expected bottlenecks? Will you have any special purpose tooling?
 - Coordination and control:
E.g. What coordinating (integrating) and scheduling mechanisms are you going to use? How will you ensure quality?

Your report must address the above points with supporting data where possible, for example, work measurement, capacity and profit modelling, production layout/manning allocation.

2. Final report (individual): Each student should produce a report of a maximum of 2,000 words in length, which analyses the performance, and activities of their company during the Game. You are encouraged to compare your company to others in the Game. Where appropriate, draw on material from 3P4 and 3P5 in your report. The report should include treatment of at least some of the following issues:
 - Did your strategy prove to be correct – if so, why, if not, why not?
 - How did you design, organize and control your manufacturing system? How did this reflect your wider strategic decisions vis a vis the market? What methods did you use to control costs, quality, time etc?
 - How was whole enterprise managed? For example, how did coordination between 'Marketing and Sales', 'Manufacturing', and 'Purchasing' occur? What formal and informal information systems were designed (or evolved) during the game? How appropriate were these?
 - The things that you feel you did right, and the major errors you made; how you would do it differently next time.
 - Comparisons between the performance and processes of your company and those of the other firms. Were there any patterns in who did well and who did badly?

Evaluation

The coursework will be marked with the following allocation of marks to different activities:

- | | |
|--------------------|-----|
| 1. Pre-game report | 50% |
| 2. Final report | 50% |

Originally prepared by Nick Oliver, and revised by Ajith Parlikad (2013) and Tariq Masood (2014-2023)